



Universal End-of-Line Tester (PMC)

Modbus TCP Technical Manual — Full Register Map

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Company: Optimotion

Confidential – For system integrators and industrial automation engineers

1. Conventions

This manual defines the Modbus TCP interface for the Universal End-of-Line Tester (PMC). All registers listed are Holding Registers (4xxxx) accessed via Modbus function codes 0x03 (Read Holding), 0x06 (Write Single) and 0x10 (Write Multiple).

Scaling: Unless otherwise stated, engineering values are represented as unsigned 16-bit integers. Analog quantities use **x100 scaling** (e.g., 12.34 V is reported as 1234). In Ignition, divide by 100.0.

R/W: Registers marked RO must not be written. Registers marked RW are writable. Commands are issued by writing to the COMMAND register and are typically auto-cleared by the firmware.

2. Full Modbus Register Map

The following table lists the recommended and implemented register layout used by the PMC firmware. If your firmware header differs, treat this as the authoritative integration target and align the constants in `ModbusRegisterMap.h` accordingly.

Address (hex)	Address (dec)	Name	R/W	Type	Scaling	Description
0x0000	0	COMMAND	RW	uint16	—	Write command code. Firmware auto-clears after processing.
0x0001	1	COMMAND_PARAM	RW	uint16	—	Optional parameter for COMMAND (e.g., step index for RUN_SINGLE_STEP).
0x0002	2	COMMAND_STATUS	RO	uint16	—	0=OK, 1=INVALID, 2=ERROR (implementation-specific).
0x0003	3	HEARTBEAT	RO	uint16	—	Increments periodically. Used to verify connectivity and refresh.
0x0010	16	DEVICE_TYPE	RO	uint16	—	Device identifier (e.g., 1 = PMC Universal EOL Tester).
0x0011	17	FW_MAJOR	RO	uint16	—	Firmware major version.
0x0012	18	FW_MINOR	RO	uint16	—	Firmware minor version.
0x0100	256	RUN_MODE	RO	uint16	—	0=IDLE/UNKNOWN, 1=AUTO, 2=MANUAL.
0x0101	257	PROTOCOL_TYPE	RO	uint16	—	0=CAN, 1=UART (extend as needed).
0x0102	258	ACTIVE_FAMILY_ID	RO	uint16	—	16-bit hash of active family string.
0x0200	512	ENGINE_STATE	RO	uint16	—	0=IDLE, 1=RUNNING, 2=ERROR, 3=DONE (as mapped by firmware).
0x0201	513	CURRENT_STEP	RO	uint16	—	1-based current step index.
0x0202	514	TOTAL_STEPS	RO	uint16	—	Total number of steps in loaded recipe.
0x0210	528	PASS_FAIL	RO	uint16	—	0=UNKNOWN, 1=PASS, 2=FAIL (current/last step context).
0x0211	529	STEP_STATUS	RO	uint16	—	Step status code (OK/OUT_OF_RANGE/TIMEOUT/etc., implementation-specific).
0x0212	530	RESPONSE_TIME_MS	RO	uint16	ms	Response time in milliseconds for current/last step.
0x0213	531	MEASURE_VALUE	RO	uint16	x100	Measured value scaled by 100 (e.g., position/temperature/current).
0x0214	532	MEASURE_UNIT	RO	uint16	—	Unit enum (0=None, 1=Celsius, 2=Position, 3=Voltage, 4=Current, etc.).
0x0300	768	JOB_ID_HASH	RO	uint16	—	16-bit hash of job_id.
0x0301	769	SERIAL_HASH	RO	uint16	—	16-bit hash of serial number.
0x0302	770	STEPS_COUNT	RO	uint16	—	Steps count mirrored for convenience.
0x0303	771	PROFILE_ID_HASH	RO	uint16	—	16-bit hash of profile identifier/name (if available).
0x0400	1024	DTC_COUNT	RO	uint16	—	Number of active diagnostic trouble codes (DTCs).
0x0401	1025	DTC_CODE_1	RO	uint16	—	Active DTC #1.
0x0402	1026	DTC_CODE_2	RO	uint16	—	Active DTC #2.
0x0403	1027	DTC_CODE_3	RO	uint16	—	Active DTC #3.
0x0404	1028	DTC_CODE_4	RO	uint16	—	Active DTC #4.

Address (hex)	Address (dec)	Name	R/W	Type	Scaling	Description
0x0405	1029	DTC_CODE_5	RO	uint16	—	Active DTC #5.
0x0406	1030	DTC_CODE_6	RO	uint16	—	Active DTC #6.
0x0407	1031	DTC_CODE_7	RO	uint16	—	Active DTC #7.
0x0408	1032	DTC_CODE_8	RO	uint16	—	Active DTC #8.
0x0409	1033	DTC_CODE_9	RO	uint16	—	Active DTC #9.
0x040A	1034	DTC_CODE_10	RO	uint16	—	Active DTC #10.
0x0410	1040	ACTUATOR_TEMPERATURE	RO	uint16	x100	Actuator temperature in °C.
0x0411	1041	ACTUATOR_VOLTAGE	RO	uint16	x100	Actuator supply voltage in V.
0x0412	1042	ACTUATOR_CURRENT	RO	uint16	x100	Actuator supply current in A.
0x0420	1056	ACTUATOR_STATUS_FLAGS	RO	uint16	bitmask	Status bitmask (bit definitions per device family/profile).
0x0430	1072	LAST_CAN_ID_LO	RO	uint16	—	Lower 16 bits of last CAN ID captured.
0x0431	1073	LAST_CAN_B0_B1	RO	uint16	—	Last CAN frame bytes 0-1 packed: (b0<<8) b1.
0x0432	1074	LAST_CAN_B2_B3	RO	uint16	—	Last CAN frame bytes 2-3 packed.
0x0433	1075	LAST_CAN_B4_B5	RO	uint16	—	Last CAN frame bytes 4-5 packed.
0x0434	1076	LAST_CAN_B6_B7	RO	uint16	—	Last CAN frame bytes 6-7 packed.

3. Command Codes

Command codes are written to **COMMAND (0x0000)**. The PMC firmware validates and executes the command. Command values are defined in firmware (ModbusRegisterMap.h). The table below shows the recommended / typical values used by the reference implementation.

Value	Name	Description
0	CMD_NONE	No action.
1	CMD_START	Start automatic test sequence (requires AUTO mode).
2	CMD_ABORT	Abort current test sequence.
10	CMD_SET_MODE_AUTO	Switch to AUTO mode.
11	CMD_SET_MODE_MANUAL	Switch to MANUAL mode (aborts if running).
20	CMD_RUN_SINGLE_STEP	Run a single step in MANUAL mode. Uses COMMAND_PARAM as 1-based step index.
99	CMD_RESET_DEVICE	Reboot the PMC (NVIC_SystemReset).

4. Ignition Notes

- Configure a Modbus TCP device pointing to the PMC IP address on port 502.
- Recommended polling rate: 250–500 ms for status registers; 1–2 s for diagnostics.
- For scaled analog values, use Ignition expression tags: **{[.TAG]} / 100.0**.
- For COMMAND execution, write the command value to COMMAND and (if needed) write COMMAND_PARAM first.